Question Number : 109 Question Id : 640653450846 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

How many unique *du-paths* are there for the variable: `x`, `y` and `z`?

Options :

6406531500494. ✔ 7

6406531500495. ✖ 8

6406531500496. ✖ 9

6406531500497. ✖ 10

Industry 4.0

Section Id :	64065329296
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	45
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065364761
Question Shuffling Allowed :	No

Is Section Default? : null

Question Number : 110 Question Id : 640653450849 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: INDUSTRY 4.0"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECT TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT ,PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531500506. ✔ Yes

6406531500507. ✖ No

Sub-Section Number : 2

Sub-Section Id : 64065364762

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 111 Question Id : 640653450863 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of them is not a part of the 7 principles of IoT:

Options :

6406531500539. ✖ Big Analog Data

6406531500540. ✔ Quality Assurance

6406531500541. ✖ Constant connection

6406531500542. ✖ Real-time

6406531500543. ✖ Spectrum of insights

Sub-Section Number :	3
Sub-Section Id :	64065364763
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 112 Question Id : 640653450861 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the below formula represents the Metropolitan metric distance?

Options :

6406531500531. ✖ $dij=(xi-xj)^2+(yi-yj)^2$

6406531500532. ✔ dij=(xi-xj)+(yi-yj)

6406531500533. ✖ dij = (x2 – x1) + (y2 – y1)

6406531500534. ✖ All of these.

Sub-Section Number :	4
Sub-Section Id :	64065364764
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450851 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (113 to 115)

Question Label : Comprehension

The ten activities specified below are to be performed to make a product. Then answer the given subquestions.

Activity	Activity Time (in min)	Preceding Activity
A1	9	
A2	5	
A3	18	
A4	10	
A5	6	
A6	8	A1, A3
A7	16	A1, A4
A8	19	A3, A5
A9	5	A2
A10	13	A8, A9

Table Indicating Activity Information

Sub questions

Question Number : 113 Question Id : 640653450852 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are an allowable grouping of activities to produce the product, if the target is to have the maximum possible output?

Options :

6406531500512. ✔ (A3)→(A1,A4)→(A2,A5,A6)→(A8)→(A9,A10)→(A7)→Output

6406531500513. ✖ (A3,A2)→(A1,A4)→(A5,A6)→(A8)→(A10,A9)→(A7)→Output

6406531500514. ✖ (A2,A5,A6) → (A1,A4)→(A3)→(A8)→(A10,A9)→(A7)→Output

6406531500515. ✖ (A3)→(A1,A4,A2)→ (A8)→(A10,A9)→(A7)→Output

Question Number : 114 Question Id : 640653450853 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the balance delay for the line (or lines) identified in the previous question? *(Enter your answer in % rounded to two decimal places, without the percentage symbol. Eg. If you compute the answer as 0.10005, then enter 10.01 as your answer)*

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 115 Question Id : 640653450854 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

If a day has 2 shifts and each shift is for 8 hours, then what is the per day maximum output that can be achieved from the layout(s) chosen as per the previous question number 5?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

50

Question Id : 640653450857 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (116 to 117)

Question Label : Comprehension

The Bellman equation is solved using backward recursion for a 3-period inventory model. The ordering cost is Rs. 2500 per order; the inventory holding cost is Rs. 250 per unit per period; the material cost is Rs. 15,000 per unit, and the item is sold at Rs. 30,000 per unit. The inventory can hold at most 30 units at any given period. Then answer the given subquestions.

Sub questions

Question Number : 116 Question Id : 640653450858 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

There are 20 units at the start of the second period and the demand in the period is 20 units. The possible future profit is Rs. 7,47,500 irrespective of how many units are in the inventory at the end of the second period. Then what is the maximum profit that can be generated up to period 2 (when the possible actions are restricted to ordering either 0, 10 or 20 units in period 2)?

(Hint: You are to choose the best profit using backward recursion) (Note: Enter the numerical answer in Rupees, without the "Rs." notation. Eg. If you get a profit of Rs. two thousand (Rs. 2000), enter the answer as 2000)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1347500

Question Number : 117 Question Id : 640653450859 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

If the demand in the second period is not deterministic and follows a distribution as given in table below, then what is the probability, that period 2 will end with 0 units in inventory if period 2 starts with 20 units in inventory and an order is placed for 20 units?

(Hint: You need to find the transition probability that period 3 will begin with 0 units of inventory if period 2 starts with 20 units in inventory and an order of 20 units is placed in period 2)

(Note: Enter your answer in decimal values only.)

Demand	Probability of Demand
10	0.5
20	0.25
30	0.125
40	0.125

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.125

Question Id : 640653450864 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (118 to 119)

Question Label : Comprehension

Let us assume a flight with capacity c =100 (identical seats), and has the price for an economy traveller to be $200, and the price for a business traveller to be $600. If the demand of business travellers is uniformly distributed between [10,30], then answer the given subquestions.

Sub questions

Question Number : 118 Question Id : 640653450865 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the protection level? (Round your answer to nearest higher integer. For example, if answer is 10.5 round it to 11)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

22 to 24

Question Number : 119 Question Id : 640653450866 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the booking limit? (Round your answer to nearest higher integer. For example, if answer is 10.5 round it to 11)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

76 to 78

Question Id : 640653450876 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (120 to 123)

Question Label : Comprehension

A company needs to ship two products (A and B) to a foreign country. Two models of transport (Air and Sea) are available for shipping. The demand in the foreign market is stable at 100 units per week for each of the two products. Products A and B cost Rs. 20,000 and Rs. 10,000 respectively. If the company chooses to use air as the mode of transport, then it must ship 100 units in every shipment. On the other hand, if Sea is chosen as the mode of transport, then the shipment size must be 400 units. Air shipments take 1 week to reach the destination while Sea shipments take 4 weeks. Items shipped by Air incur a cost of Rs. 360 per unit and those shipped by Sea incur Rs. 90 per unit. If the company incurs 20% of the per unit cost for a product as the cost to hold the item in inventory for a year, answer the given subquestions. (Assume that a year has 52 weeks)

Sub questions

Question Number : 120 Question Id : 640653450877 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

When shipping Product-A by Air, what is the total inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

150

Question Number : 121 Question Id : 640653450878 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

When shipping Product-A by Sea, what is the total inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

600

Question Number : 122 Question Id : 640653450879 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the total cost that is incurred for the movement of Product-B by Air (enter your answer in Lakh of rupees without the rupee symbol and punctuations. E.g: If you get an answer as “1 Lakh”, then enter 100000)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2172000

Question Number : 123 Question Id : 640653450880 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the total cost that is incurred for the movement of Product-B by Sea (enter your answer in Lakh of rupees without the rupee symbol and punctuations. E.g: If you get an answer as “1 Lakh”, then enter 100000)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1668000

Sub-Section Number :	5
Sub-Section Id :	64065364765
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450867 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (124 to 125)

Question Label : Comprehension

During COVID, a company had asked its salesman to visit a set of locations (L1 to L5) to carry out the business. Now, post-COVID, business is picking up. Hence, the company is planning on hiring more salesmen. However, all salesmen must come to the office (OF) every day before leaving for the different locations, and return back to the office (OF) at the end of the day. Given the distance matrix for the locations set as below, answer the given subquestions.

Location	L1	L2	L3	L4	L5
L1					
L2	2				
L3	9	6			
L4	8	5	9		
L5	10	6	8	6	
OF	4	2	5	2	5

Distance Matrix in Kilometres

Sub questions

Question Number : 124 Question Id : 640653450868 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Then, during COVID, which of the following route(s) would have been best for the salesman?

Options :

6406531500546. ✔ L1 → L3 → L4 → L2 → L5 → L1

6406531500547. ✔ L1 → L4 → L3 → L2 → L5 → L1

6406531500548. ✔ L2 → L3 → L4 → L1 → L5 → L2

6406531500549. ✖ None of these

Question Number : 125 Question Id : 640653450869 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Post-COVID, if a salesman should not travel more than 10 Kilometers in a day, then how many salesmen will have to be hired by the company to visit all the locations every day?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

Sub-Section Number :	6
Sub-Section Id :	64065364766
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653450872 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (126 to 128)

Question Label : Comprehension

200 units of an LCD TV are sold every day in a store. Each LCD TV is produced at a cost of Rs. 500. A fixed cost of Rs. 1500 is incurred for a lot (irrespective of the number of items in the lot) to produce it. 30% of the cost to produce the LCD TV is incurred to store the LCD TV in the inventory for one year. If the LCD TV is sold for 365 days a year.

Based on the above data, Answer the given subquestions.

Sub questions

Question Number : 126 Question Id : 640653450873 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the economic order quantity for the LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

1208

1209

Question Number : 127 Question Id : 640653450874 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the average inventory that must be carried for the LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

604

605

Question Number : 128 Question Id : 640653450875 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

How many orders will be placed for LCD TVs?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

60

61

Sub-Section Number :	7
Sub-Section Id :	64065364767
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 129 Question Id : 640653450856 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

What are the inputs for an order-upto-level calculation? (Choose all that are applicable)

Options :

6406531500522. ✔ Sales forecast

6406531500523. ✔ Lead time

6406531500524. ✔ On-hand inventory

6406531500525. ✔ Safety stock

Question Number : 130 Question Id : 640653450862 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are correct? (choose all that is applicable)

Options :

6406531500535. ✔ Set covering problem minimizes the number of facilities located under the constraints that all demand needs to be served

6406531500536. ✔ Some budgetary constraints might not allow us to serve all the demand locations

6406531500537. ✔ We locate the budgeted facilities to maximize the service demand call “maximal covering location problems"

6406531500538. ✖ None of these

Sub-Section Number :	8
Sub-Section Id :	64065364768
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 131 Question Id : 640653450850 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Match the layout to the possible production system it may have *(Hint: A "Layout" can be possible for two or more "Production Systems". Hence, choose the options carefully)*

Layouts

i. Product Layout

ii. Process Layout

iii. Fixed Position Layout

Production Systems

A. Continuous production (identical items are produced in very large quantities)

B. Mass production (similar items are produced in large quantities)

C. Batch production (identical items are produced in groups. Different item groups have different characteristics)

D. Job-Shop production (Items produced are unique)

Options :

6406531500508. ✖ (i) – A, B; (ii) – C, D; (iii) – D

6406531500509. ✔ (i) – A, B, C; (ii) – C, D; (iii) – D

6406531500510. ✖ (i) – A; (ii) – B, C; (iii) – D

6406531500511. ✖ (i) – A, B, C; (ii) – D; (iii) – D

Question Number : 132 Question Id : 640653450860 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are False? Choose all those that are applicable

Options :

6406531500528. ✖ Euclidean distance measures distance as the straight line between 2 points

6406531500529. ✖ Metropolitan distance measures distance with respect to the road availability

6406531500530. ✔ None of these

Question Number : 133 Question Id : 640653450870 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is Digitalization? Choose all that are applicable

Options :

6406531500551. ✖ Conversion of physical 2D drawings to CAD drawings

6406531500552. ✔ Creating PLC logic in a microprocessor-based system to process data

6406531500553. ✖ Providing real-time maintenance schedule based on real-time machine performance

6406531500554. ✖ None of these

Question Number : 134 Question Id : 640653450871 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

What is Digitization? Choose all that are applicable

Options :

6406531500555. ✖ Use of digital technologies to change the processes

6406531500556. ✖ Process of moving towards a digital business

6406531500557. ✔ Pure Analog to Digital Conversion

6406531500558. ✔ Mere transformation – encoded in digital format

Sub-Section Number :	9
Sub-Section Id :	64065364769
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 135 Question Id : 640653450855 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Choose the correct term from the Column B for the description given in Column A. Note that one to-one mapping in the table is not necessary

Column A	Column B
a.1 A customer can book a product on Amazon, but can collect the product from the showroom	b.1 Impulse buying
a.2 You are waiting at the cash counter of a shopping mall. While waiting, you could see chocolates lying in the adjacent shelves. So you picked a couple of those.	b.2 Multi-channel retailing
a.3 You picked up tea packets to your trolley while shopping at the shopping mall. You could see sugar and coffee packets also on the nearby shelves. You suddenly remember that sugar is out of stock at home. So you picked up a couple of sugar packets. So, Tea and Sugar are---	b.3 Complementary product
	b.4 Omni-channel retailing
	b.5 Substitution product
	b.6 Inventory-dependent demand

Options :

6406531500518. ✔ a.1: b.4, a.2:b.1, a.3:b.3

6406531500519. ✖ a.1: b.1, a.2:b.2, a.3:b.3

6406531500520. ✖ a.1: b.3, a.2:b.1, a.3:b.4

6406531500521. ✖ None of these